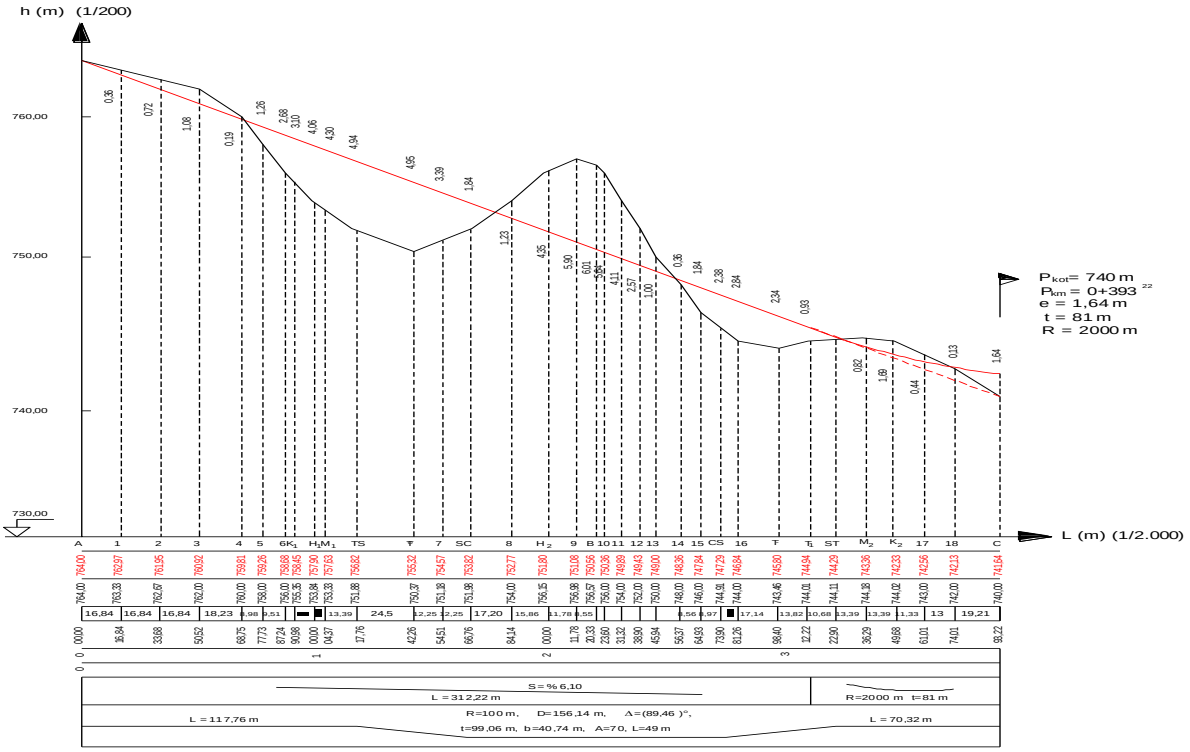
8. PROFILE DESIGN AND VERTICAL CURVES

8.1 Profile

The section taken through the highway axis is named the **profile**.

The highway axis on the plan (the projection of the highway) is carried onto the vertical plane by using the cross-section points to obtain the profile, as seen in the figure.



The **(black) elevations** of the cross-sections are read from the contour map and marked on the vertical plane to be connected with a line. The line obtained this way shows the situation of the land and is named the **black line**. As the black line is a broken line, vehicles cannot move on it with the desired comfort and safety level. To obtain a more suitable rolling surface, the primary compulsory (obligatory) points (start and end points of the highway) are connected with each other to obtain the **red line**. The red line is sometimes a single line having a constant grade, while other times it consists of line segments with proper length, having various grades. The elevation of any point on the red line is named the **red elevation**. If the red elevation is lower than the black elevation of any point cut, else fill has to be made. If the grade of the red line of the profile is positive (+), this section of the highway is an ascending section; else, if the grade is negative (-), this section of the highway is a descending section. The curves placed between red lines are named the vertical curves.

8.1.1 Grade

The longitudinal grade given through the highway axis is shortly named the **grade**. As a highway is being designed, the grade should not pass a specific value to provide safety, comfort,

and capacity. Highway class, situation of the land, climate, dominant vehicle type, and project standards are effective in determining the highest grade to be applied. Low grades are applied on high-standard highways. The maximum grades accepted by the General Directorate of Highways are 7% on first-class highways, 8% on second-class highways, and 9% on third-class highways. 15% of grades are accepted on village roads, while on expressways, the maximum grade is 4%. The minimum value of the grade is determined to be 0,35% by the General Directorate of Highways. However, 0,5% is applied practically.

8.1.2 Drawing the red line on the profile

The red line is drawn with specific rules on the profile. The rules to be taken into account can be summarized as:

- The red line has to be drawn in a way that minimizes the earthwork and, if that is not possible, equalizes the cuts and the fills through the route.
- The sight distances have to be provided on the vertical curves that are to be designed.
- To decrease the vehicle operating costs and to increase safety, high-graded sections have to be as short as possible.
- Straight sections have to pass from a higher elevation than the natural land to avoid floods or snow drifts.
- On riverside passes, the route has to pass from a higher elevation than the highest water level.
- To ease the surface drainage, sag vertical curves have to be avoided on cut cross-sections.
- Horizontal curves and the vertical curves should not intersect with each other.
- Long straight sections with a constant grade have to be avoided in order to provide safe driving.
- At-grade junctions should not be placed on high-graded sections.
- At-grade junctions should not be placed on vertical curves.

8.1.3 Additional highway elements used on graded sections

Additional highway elements are needed in order to provide safety, comfort, and capacity on high-grade highway sections.

In this regard, climbing lanes and emergency escape ramps are constructed.

8.1.3.1 Climbing Lanes

Especially on two-lane two, two-direction highways, because of the heavy vehicle amount, the capacity decreases on graded sections. As the high vehicles move with slower velocities on graded sections, this affects the other vehicles behind them, and queuing occurs. Queuing forces impatient drivers to make mistaken maneuvers. To avoid this negative situation, climbing lanes are built on the right side of the highway on ascending sections. Experiences show that over the ascending grade of 3%, heavy vehicles generate negative effects. For this reason, climbing lanes are built when the following situations occur:

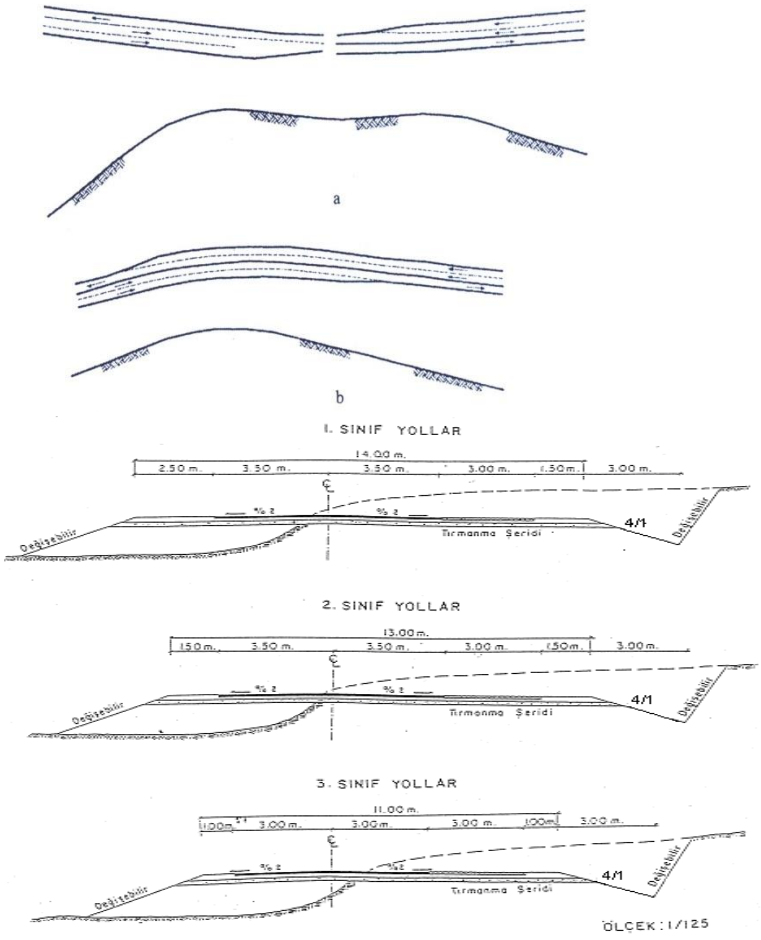
a- The traffic volume is over 200 vehicles/h, and the heavy vehicle ratio is over 20%.

b- The velocity of the heavy vehicles decreases by about 15-20 km/h because of the ascending section.

c- The traffic flow is around E and F levels on the ascending section.

d- The difference between the level of services between the section before and the graded section is two or more (Such as C-E, D-F).

Climbing lane length and width have to be calculated in order to avoid the negative effects listed previously. Climbing lanes are designed separately for each direction. However, on crest vertical curves, the climbing lanes on both sides can be planned separately (Fig. Left-a) as well as together (Fig. Left-b). The cross-sections of the climbing lanes may vary due to the highway class (Fig. Right).

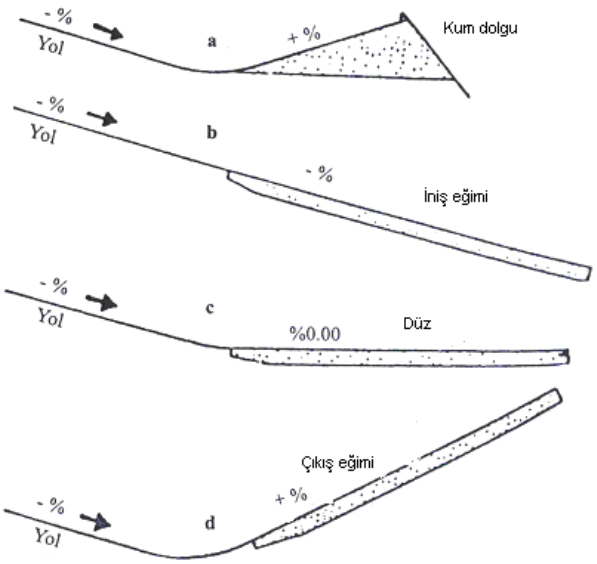


8.1.3.2 Emergency escape ramps

On high-graded descending sections, emergency escape ramps are constructed in order to provide safety. They are used especially when a brake problem occurs in order to provide an emergent and safe stop.

Emergency escape ramps are needed, especially when the brake is disabled due to overloading, a mechanical failure, or not making a proper gear shift, and the control of the vehicle is lost on high-graded descending sections. Emergency escape ramps provide the driver with an acceptable deceleration possibility and allow drivers to control their vehicles without causing any harm to themselves, the vehicle, or the environment. They are arranged as additional lanes, as shown in the figure.

Emergency escape ramps are constructed where the vehicles that lose their control cause serious accidents and are on an aligned section.



8.2 Vertical curves

The curves placed between different graded consecutive red line segments are named the **vertical curves**. The curves placed as vertical curves are **circle arcs and 2nd-degree parabolas,** independent of being crest or sag. The first group is named the circular vertical curve, and the second group is named the **parabolic vertical curve**. As the horizontal and vertical scales are different on the profile, vertical curves can not be drawn using a compass.

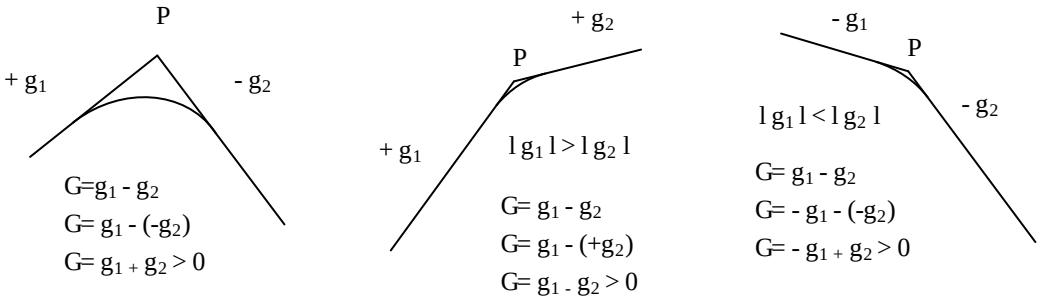
The red elevations for the cross-sections on the vertical curve have to be calculated separately. The type of the vertical curve is determined with respect to the change of grade between the consecutive red line, or in other words, according to the algebraic difference between the grades. If the angle between the first red line and the horizontal line is α, tg α =g1; If the angle between the first red line and the horizontal line is β, tgβ=g_2.

According to the algebraic difference below, a classification is made as a **crest** or **sag** vertical curve.

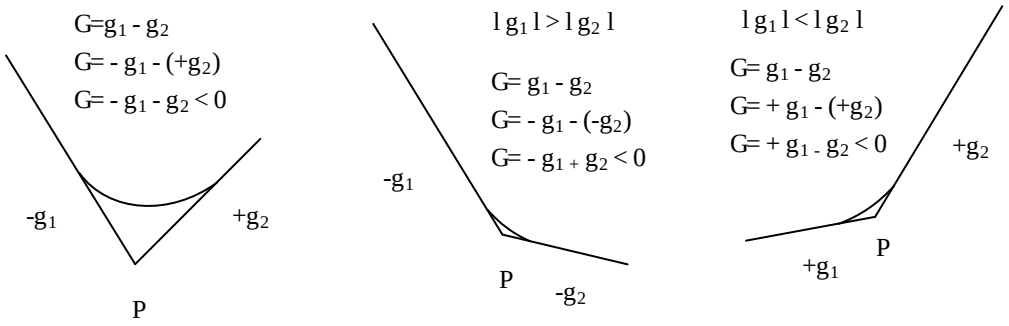
$$G = g_1 - g_2$$

On crest vertical curves, the algebraic difference is positive (+), while on sag vertical curves, the algebraic difference is negative (-). If the absolute value of the algebraic difference between the two consecutive red lines is more than 0,5%, a vertical curve is placed. On low-standard roads, as comfort is not a priority, if the algebraic difference is below 1%, a vertical curve may not be placed. On application, there are 6 types of vertical curves according to the size of g_1 and g_2 grades. Three of them are crests, and the other three of them are sag vertical curves.

Crest vertical curves



Sag vertical curves



8.2.1 Sight distance on vertical curves

In highway design, safe driving conditions and safe sight distances also have to be provided. The sight distance problem arises on the section where the highway changes its direction or, in other words, at horizontal curves on the horizontal geometry, while on vertical geometry, it arises where the consecutive grade difference occurs or, in other words, where the grade of the red line changes. Especially at crest vertical curves, the sight distance decreases as the algebraic difference between the consecutive grades increases.

The sight distance is taken into account, while the length of the vertical curve is determined and calculated differently according to whether the highway is separated or not. On **undivided highways,** the most inappropriate condition arises when a driver overtakes a vehicle in front according to the possibility of a vehicle approaching from the opposite direction.

So, the sight distance for the condition of a vehicle is approaching from the opposite direction should be taken into account for the minimum crest vertical curve length.

$$S = L_S = \frac{(d_1 + d_2)}{(v_1 \quad v_2)}.(v_1 + v_3)$$

As there isn't any vehicle approaching from the opposite direction on a divided highway, the necessary brake safety distance to avoid crashing with an obstacle is usually taken into account.

$$S = l_{bs} = 0{,}278.V.t_r + 0{,}00394.\frac{V^2}{(f \pm g)}$$

On the other hand, on undivided highways, overtake maneuvers can be made while there isn't any vehicle approaching from the opposite direction. To take this situation into account, the overtake distance has to be calculated as:

$$S = l_o = \frac{(d_1 + d_2)}{(v_1 - v_2)}.v_1$$

and the design should be made according to the most inappropriate conditions. In other words, S = max(l_{fe}, l_o) should be taken.

On the calculations above, for passing sight distance the velocity of the overtaking vehicle and for the stop sight distance the velocity of the braking vehicle is accepted as the project velocity to reflect the worst condition. Also, for passing sight distance, the velocity of the overtaken vehicle can be taken as 15 km/h lower than the overtaking vehicle. On the other hand, driver sight height can be taken as 1-1,4 m while the possible obstacle height can be taken 0,25 at maximum.

8.2.2 Comfort at vertical curves

While the vehicles are moving on the vertical curve, they are affected by a vertical acceleration that creates discomfort, just like the horizontal curves. The value of this acceleration should not exceed a specific value according to comfort. The experimental studies show that the upper limit of the bearable vertical acceleration is around 1/20 of the gravitational acceleration. This value is around 0,5 m/sec^2.

The vertical acceleration that arises on a vertical curve can be formulated as:

$$a = \frac{v^2}{R}$$

where v is the velocity of the vehicle (m/sec) and R is the curve radius (m).

If the velocity of the vehicle is written as km/h, the acceleration becomes:

$$a = \frac{V^2}{3{,}6^2 \cdot R} = \frac{V^2}{12{,}96 \cdot R} \cong \frac{V^2}{13 \cdot R}$$

Hence, the radius that provides the comfort criteria on a circular vertical curve can be found as:

$$R = \frac{V^2}{13 \quad a}$$

On a circular vertical curve, if the algebraic difference between the grades of the consecutive red lines is G=g_1-g_2 and the central angle of the circle arc is Δ=α+β by the length of the vertical curve can be written as taking Δ tg Δ = G:

$$L = R \quad \Delta = R \quad G$$

As the values of α and β angles are very small, this assumption gives realistic results.

If the radius is written in the formula, the length of the vertical curve is found as:

$$L = \frac{V^2}{13\ a}\ G$$

As the vertical acceleration is taken around 0,3~0,5 m/sec^2 interval, the length calculated this way becomes too small compared to the length calculated with respect to the sight distance. Therefore, the comfort problem almost does not exist on vertical curves that have been designed with respect to the sight distance.

8.2.3 Parabolic Vertical Curves

The second-degree parabola curves used between the consecutive red lines are named the **parabolic vertical curves**.

Properties of parabolic vertical curves have to be known for their design and application.

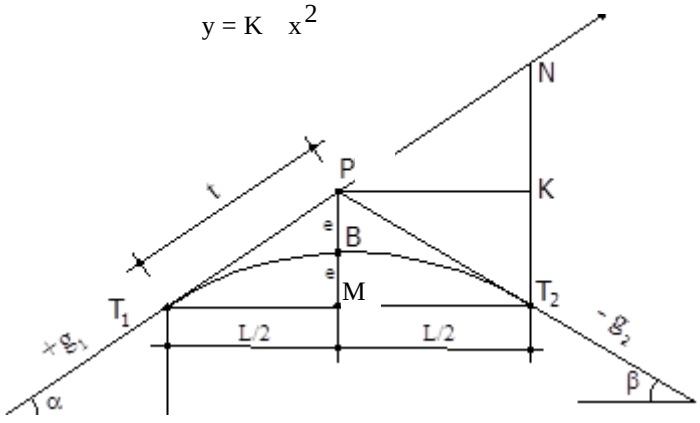
The starting point of a vertical curve is shown as **T_1**, the ending point as T2, the intersection points of the red lines as P, the external distance as e, and the length as **L**.

8.2.3.1 Properties of parabolic vertical curves

1. On vertical geometry, the angle between the red lines and the horizontal plane is very small. For example, if the angle between the 10% graded section and the horizontal plane is calculated, $\alpha=(5,71)°$, a really small value is obtained. For this reason, all calculations of the vertical curve can be assumed to be made on the horizontal plane. By using this assumption, $|T_1\ C\ T_2| = |T_1\ B\ T_2| = |T_1\ C\ T_2|$ can be written.

2. If the T_1 is taken as the origin of the vertical curve, the red line on which the T_1 is located is the x-axis, and the parallel to |P M| drawn from T_1 as the y-axis, a parabola with the K coefficient can be stated as

$$y = K\ \ x^2$$



3. The parabola passes from the center of the line that connects T1 with the M, the center point of the T_1T_2 line. In other words,

$|P\ B| = |P\ M|/2 = |B\ M|$

can be written. This distance is named the **e**, external distance.

4. The distance of the points on the parabola to the tangents, or in other words, the diversions, are symmetrical for each half of the curve with respect to the P. If $|g_1| = |-g_2|$, $T_{1_{elv}} = T_{2_{elv}}$. Else, $T_{1_{elv}} \neq T_{2_{elv}}$.

5. Unless otherwise stated, the half-length of the vertical curve is symmetrical with respect to P. Thus $|P\ T_1| = |P\ T_2|$, $|B\ T_1| = |B\ T_2|$, $|M\ T_1| = |M\ T_2|$ can be written.

On parabola, if the coordinates of the B is written by taking x=L/2 and y=e

$$e = K \cdot \left(\frac{L}{2}\right)^2 = K \cdot \frac{L^2}{4} \qquad K = \frac{4e}{L^2}$$

are found. If this is written again on the parabola equation,

$$y = \frac{4e}{L^2} \cdot x^2$$

is obtained. By using the figure,

$$|N\ T_2| = |N\ K| + |K\ T_2|$$

can be written. If $|N\ K|$ and $|K\ T_2|$ are stated in terms of red line grade and the vertical curve length,

$$\left|NK\right| = g_1 \cdot \frac{L}{2} \qquad \left|KT_2\right| = -g_2 \cdot \frac{L}{2}$$

are found and

$$\left|N\ T_2\right| = (g_1 - g_2)\ \ \frac{L}{2}$$

is obtained. By using the similarity of (T_1PM) and (T_1 N T_2) right triangles,

$$|N\ T_2| = 2|P\ M|$$

can be written. Thus,

$$\left|PM\right| = \frac{\left|NT_2\right|}{2} = \frac{(g_1 - g_2) \cdot L}{4}$$

If the calculated |PC| length is written on the formula above,

$$\left|PB\right| = \frac{\left|PM\right|}{2} = \frac{(g_1 - g_2) \cdot L}{8} = \frac{G \cdot L}{8}$$

are obtained. And this is the external distance of the vertical curve, E.

$$e = \frac{G \cdot L}{8}$$

By writing the parabola coefficient, K, in different forms, the equations below are obtained.

On vertical curve calculations, the accuracy of the parameters can be checked by using at least two of these formulas.

As $y = K \cdot x^2$, $e = \frac{G \cdot L}{8}$, $t = \frac{L}{2}$ various forms of the K equation are,

$$K = \frac{4e}{L^2} = \frac{4.\left(\frac{G.L}{8}\right)}{L^2} = \frac{G}{2.L}$$

$$K = \frac{G}{2.L} = \frac{G}{2.(2t)} = \frac{G}{4t}$$

$$K = \frac{4e}{L^2} = \frac{4.e}{(2t)^2} = \frac{4.e}{4t^2} = \frac{e}{t^2}$$

$$y = \frac{4e}{L^2}.x^2 = \frac{G}{2L}.x^2 = \frac{G}{4t}.x^2 = \frac{e}{t^2}.x^2$$

8.2.3.2 Calculations of parabolic vertical curves

The calculation of parabolic vertical curve length varies according to a crest or a sag vertical curve. By using the lengths obtained in these calculations, km, and elevations of some critical points on the curve can be calculated.

The critical points on the vertical curve calculation are the start of the curve, PVC, the end of the vertical curve, PVT, and the middle point of the curve, B. The turning point (TP), which will be discussed later on, is also critical. If the km and elevations of the PVI are known, the km and elevations of the critical points can be calculated as:

Km Calculations:

$T_{1_{km}} = P_{km} - t$

$B_{km} = P_{km}$

$T_{2_{km}} = P_{km} + t$

Elevation Calculations :

$T_{1_{elv}} = P_{elv} - g_1.t$

$B_{elv} = P_{elv} - e$

$T_{2_{elv}} = P_{elv} + g_2.t$

When the elevations of the tangent points are being calculated, it is important to write the signs of the relevant red line grades correctly. In other words, if the g_1 grade is ascending, it takes the

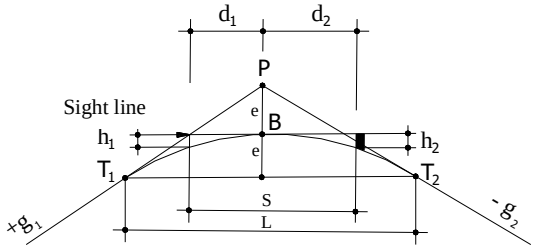
+ sign; otherwise, it takes the – sign. The same is valid for the g_2 grade. When the elevation of the B point is being calculated, the sign of the algebraic difference between the grades (G) is used similarly. In other words, G takes a + positive sign on crest vertical curves and a – negative sign on sag vertical curves. For calculations, it should not be forgotten to use G as G/100.

8.2.3.2.1 Parabolic crest vertical curves

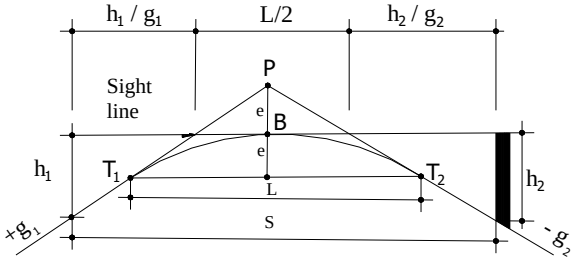
There are two different conditions for the calculation of the minimum length of these types of curves.

At first condition, the sight distance is shorter than the curve length (inside the curve); also, the length of the potential obstacle is short (Fig. a).

In the second condition, the sight distance is longer than the curve length (outside the curve), and the length of the obstacle is long (Fig. b).



a- S<L



b- S>L

By using the formulas for the given two conditions, the curve length is found according to the one that provides the S sight distance. For this reason, S is placed in the inequalities and compared with L. Only one of the inequalities provides the sight distance. In application, both formulas are used to see the situation clearly, and then an elimination is made for control.

For S < L: $$L = \frac{G \quad S^2}{(\sqrt{2h_1} + \sqrt{2h_2})^2}$$

For S > L: $$L = 2S - \frac{2(\sqrt{h_1} + \sqrt{h_2})^2}{G}$$

If the stop-sight distance is taken into account on crest vertical curve calculations, driver sight height h_1=1,14 m and the potential obstacle height h_2=0,15 m that are advised by AASHTO And accepted by the General Directorate of Highways are written on the formula,

For S < L: $$L = \frac{G \quad S^2}{4,2}$$

For S > L: $$L = 2S - \frac{4,2}{G}$$

are obtained. If the vertical curve calculations are to be made for passing sight distance, h_1=1,14 m and h_2=1,37 m are taken, and the formulas become:

For S < L: $L = \frac{G \quad S^2}{10}$

For S > L: $L = 2S - \frac{10}{G}$

On calculations, algebraic differences should be taken as G/100. Crest vertical curve lengths should not be lower than 120 m on state highways and 80 m on provincial roads due to comfort. If lower values are obtained on calculations, minimum values for the comfort criterion are taken for calculations.

8.2.3.2.2 Parabolic sag vertical curves

There isn't any sight problem on the day driving for this type of curve unless there's an overpass or a similar obstacle on the route. However, the effect of headlamp lights should be considered when driving at night. For this reason, it is important to know the stop-sight distance under the effect of headlamp lights. As the distance of the headlights to the rolling surface is h and the vertical deflection of the headlamp light beams from the vehicle axis is β, the equations used for vertical curve length are:

For S < L: $L = \frac{G \quad S^2}{2(h + S \quad tg\beta)}$

For S > L: $L = 2S - \frac{2(h + S \quad tg\beta)}{G}$

If h=0,61 m and β=1° are accepted, the equations become:

For S < L: $L = \frac{G \quad S^2}{1,22 + 0,035 \quad S}$

For S > L: $L = 2S - \frac{1,22 + 0,035 \quad S}{G}$

If a highway route passes under a railway or another highway, the profile falls into the sag vertical curve. The sag vertical curve is designed to provide the minimum sight distance. So, two conditions are possible with respect to the sight distance being shorter or longer than the vertical curve length. The following equations are used for calculations. As the driver sight height is h_1, the potential obstacle height h_2, and free height of the overpass is H,

For S < L: $L = \frac{G \quad S^2}{8(H - \frac{h_1 + h_2}{2})}$

For S > L: $L = 2S - \frac{8}{G}(H - \frac{h_1 + h_2}{2})$

The General Directorate of Highways advises the use of the following equation: H=5 m, h1= 1,83 m, and h_2 = 0,46 m.

For S < L: $L = \frac{G \quad S^2}{32,5}$

For S > L: $L = 2S - \frac{32,5}{G}$

8.2.3.2.3 Vertical curve calculations using the curve coefficient

The velocity of the grade change on the curve should be within the acceptable limits for drive comfort. For this reason, the curve length should be sufficient. It is also important to provide a good sight distance on the parabolic vertical curve. The absolute value of the vertical curve coefficient has a similar function with a radius of a circle and is a criterion of the comfort, safety, drainage, aesthetics, and similar properties that a vertical curve provides. According to this equation, K can also be described as the curve length per unit grade difference. The minimum K values that provide the stop sight distance and passage sight distance of a vertical curve are given in the table.

$$K = L / G$$

K : Vertical curve coefficient

L : Parabolic vertical curve length (m)

G : Algebraic difference of the grades

Vertical curve coefficient values

Project Velocity (km/h)	For Stop Sight Distance			For Passage Sight Distance	
	SSD (m)	Crest Vertical Curve K	SagVertical Curve K	SSD (m)	Crest Vertical Curve K
20	20	1	3	-	-
30	30	2	4	200	46
40	50	6	9	270	84
50	65	10	13	345	138
60	80	14	16	410	195
70	100	23	22	485	272
80	125	35	28	540	338
90	150	51	35	615	438
100	180	73	44	670	520
110	210	100	52	730	617
120	240	130	60	775	695
130	270	165	69	815	769

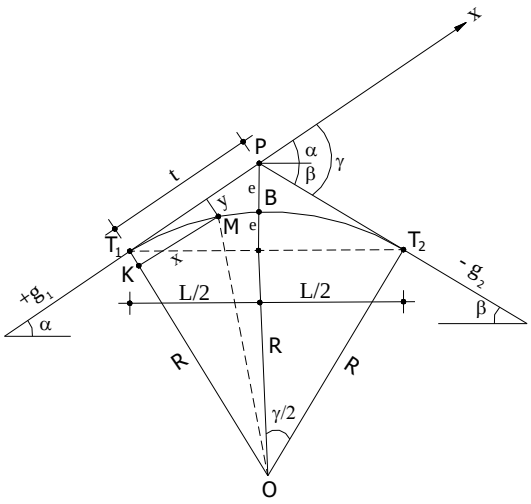
8.2.4 Circular vertical curves

Another type of curve that can be placed between the red lines is a circle arc.

Vertical curves designed using a circle with a radius of R are named **circular vertical curves**.

8.2.4.1 Properties of the circular vertical curves

The same assumptions used on parabolic vertical curve calculations are valid for circular vertical curve calculations. Also, the figure on the right can be used due to the usage of a circle with a radius of R.



According to the figure, $\gamma = \alpha + \beta$

can be written as well as $tg\,\gamma = tg(\alpha + \beta)$

If this equation is written as the trigonometric sum equation, it is written as:

$$tg\,\gamma = \frac{tg\alpha + tg\beta}{1 - tg\alpha \quad tg\beta}$$

As α and β values are too small, the "tgα.tgβ" term can be neglected. So the sum equation becomes,

$$tg\,\gamma \qquad tg\alpha + tg\beta = g_1 + g_2$$

or, in other words $\gamma \quad tg\,\gamma = G$

The length of the circular vertical curve is just like the horizontal curves can be found as,

$$t = R \quad tg(\frac{\gamma}{2})$$

If $tg(\frac{\gamma}{2}) = \frac{G}{2}$ is taken, $t = R \quad \frac{G}{2}$ is obtained.

As L=2.t is known, the length of the circular vertical curve is found as:

$$L=G.R$$

To obtain the external distance of the circular vertical curve, the O T_1 P right triangle can be used and $(e+R)^2 = t^2 + R^2$ can be written and

$$e^2 + 2Re + R^2 = t^2 + R^2 \qquad e^2 + 2Re = t^2$$

are obtained. Here, the e^2 term can be neglected when compared to the others, so the formula ultimately becomes, $2Re = t^2$

By using this formula, **e** and **2R** can be written as,

$$e = \frac{t^2}{2R} \qquad 2R = \frac{t^2}{e}$$

Another right triangle on the figure (C K M triangle) is obtained by connecting an M point taken in the curve, the perpendicular taken from this point to the radius of the circle tangent to the PVT and the center point of the curve. By using this triangle,

$$|OK|^2 + |KM|^2 = |OM|^2$$

Equations can be written and,

$$(R - y)^2 + x^2 = R^2$$

are obtained. If the equation is written in an open form, it becomes,

$$R^2 - 2Ry + y^2 + x^2 = R^2$$

$$2Ry - y^2 = x^2$$

Here, y^2 term can be neglected between the others and

$2Ry = x^2$ $\quad y = \frac{x^2}{2R}$ Equations are found. As the equation of a circle arc is known to be $y=K.x^2$, circle constant (in other words, circular vertical curve constant) is found as

$$K = \frac{1}{2R}$$

On the other hand, if instead of the 2R term, the (t^2/e) term is written, the formula of a circle can be stated as: $y = \frac{e}{t^2} \; x^2$

Here, another statement of the circle constant can be obtained as: $K = \frac{e}{t^2}$

Finally, if the $t = R \; \frac{G}{2}$ term is written on $e = \frac{t^2}{2R}$,

$$e = \frac{R^2 \quad G^2}{4 \quad 2R} = \frac{R \quad G^2}{8}$$

is obtained. As already stated before, the length of the circular vertical curve is L=R.G. If it is written on the formula above, the external distance of the circular vertical curve can be found as:

$$e = \frac{G \quad L}{8}$$

It can be seen that the external distance of a circular vertical curve is the same as that of a parabolic vertical curve. On circular vertical curves, sight distances are not verified like on the parabolic vertical curves. Instead, minimum curve radii, which will satisfy the sight distances, are given. If the advised curve radius is selected, sight distance is also provided. However, if the selected curve radius is lower than the advised value, sight distances have to be verified. The advised minimum vertical curve radii can be seen in the table.

Vertical Curve Type and the Terrain Type	**R_{min} (m) according to the highway type**			
	I	**II**	**III**	**IV**
Crest Vertical Curve				
Flat	25.000	15.000	10.000	5.000
Undulating	15.000	10.000	5.000	2.500
Mountainous	5.000	2.500	1.500	1.000
Sag Vertical Curve				
Flat	8.000	5.000	3.000	2.000
Undulating	5.000	3.000	2.000	1.500
Mountainous	2.000	1.500	1.200	1.000

8.2.5 Vertical curve calculations using the table

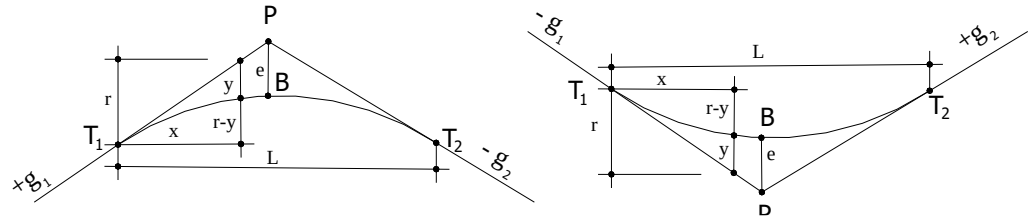
Vertical curve calculations are made by using tables in order to control the elevation and kilometer values obtained for characteristic points and red elevations of the other cross-sections. The first row of the table belongs to the start (T_1), and the last row of the table belongs to the end (T_2) cross-section. The first red line with a grade of g1 is accepted as the x-axis. So, as T_1 is the start of the vertical curve, for any cross-section on the vertical curve, the abscissa and ordinates are x and y. On the bisector point, x=L/2 and y=e. If the perpendicular distance of the projection on the axis of any point on the vertical curve to the start of the curve is shown by **r**,

$$r=g_1.x$$

can be calculated. The sign of the g_1 becomes positive (+) if the red line is ascending or negative (-) if the red line is descending. The red elevation of any point on the vertical curve can be calculated as

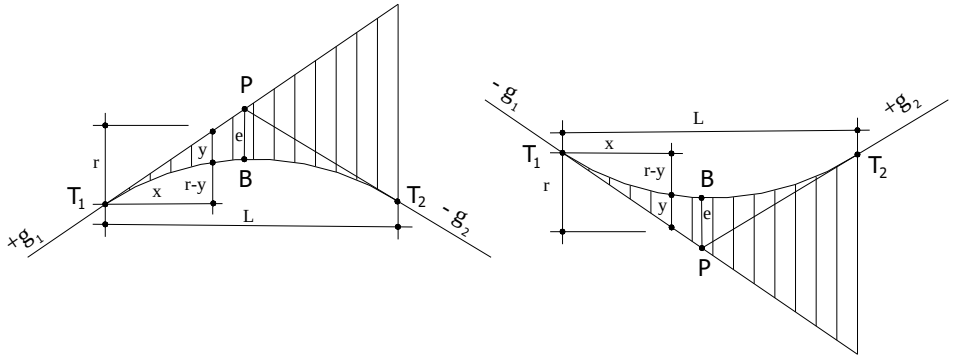
$$h_{re}=T1_{elv}+r\pm y$$

On crest vertical curves, as the algebraic difference of the grades is always positive (+), the sign of y is always negative (-); on sag vertical curves, as the algebraic difference of the grades is always negative (-), the sign of y is always positive (+).



It should not be forgotten that the calculations are made by taking the grade of the first red line (g_1) and the start point (T_1) into account, and the curve is defined with a second-degree equation.

Thus, the calculated red elevations change the direction at a specific point and start to decrease or increase. This point is known as the turning point, and its location and the elevation have to be calculated.



The calculations are followed by the order of columns. The accuracy of the calculations can be verified as:

At T_1, $T1_{elv}+r\pm y = T1_{elv}$.

At bisector cross-section, $x = t = L/2$; $T1_{elv}+r = P_{elv}$, $y = e$; $T1_{elv}+r\pm y = B_{elv}$.

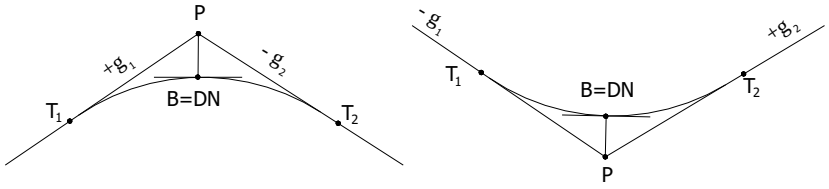
At T_2, $x=L$, $T1_{elv}+r\pm y = T2_{elv}$.

The bold cells in the table below should give the values obtained by length and elevation calculations for characteristic points. Otherwise, there's a mistake in the calculations, and it has to be checked. If the calculations are true, the deviation between the elevation values can be at the mm level due to rounding. So, the deviation at the cm level shows a mistake.

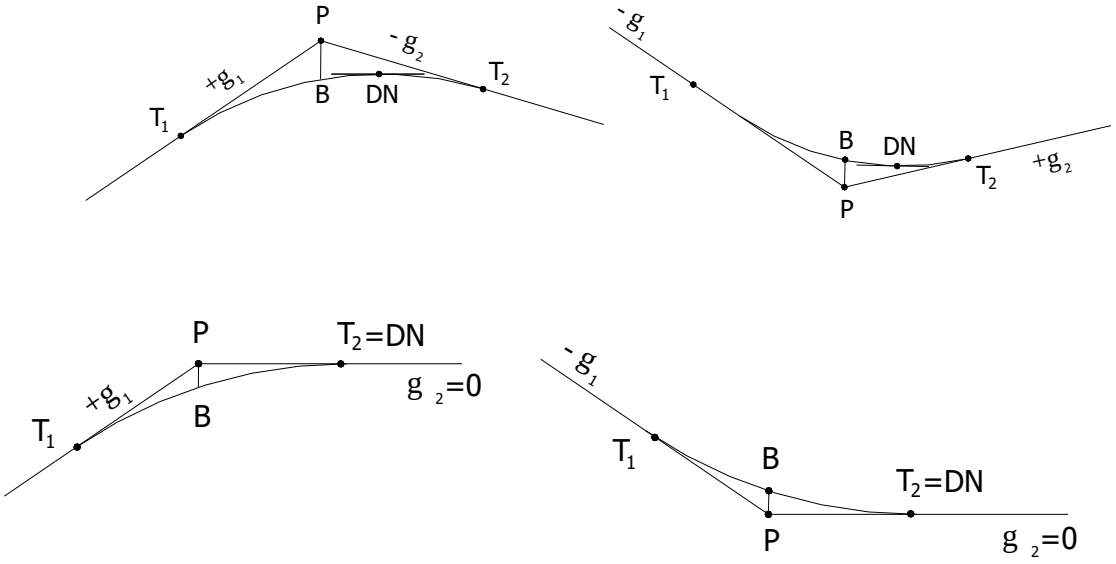
Cross-section	Km	x (m)	$r = g_1.x$	$PVC_{elv}+r$	$y=K.x^2$	$PVC_{elv}+r\pm y$
T_1						**PVC_{elv}**
B		**x=t**		**P_{elv}**	**y=e**	**B_{elv}**
T_2		**x=L**				**PVT_{elv}**

8.2.5 Calculation of turning point at vertical curves

The point having the maximum or minimum elevation on the vertical curve is named the **turning point (TP-DN)**. On this point, the line tangent to the curve is horizontal. If the absolute value of the grades of the red lines is equal to each other, the bisector point is the turning point. As the derivative of the curve equation is zero, the turning point shows a minimum or a maximum.



If the red lines are not symmetrical according to the P, the turning point falls outside the perpendicular drawn from the bisector point to the horizontal (P B line). The place of the turning point is on the lower graded line because as the grade of the red line approaches the horizontal, the place of the turning point approaches the tangent point. In the limit case, if one of the red lines is horizontal, the turning point coincides with the tangent on this line and corresponds to the same point.

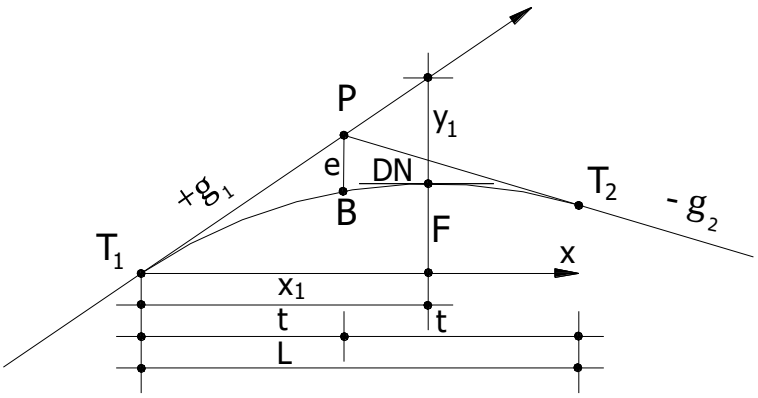


Calculation of the kilometer and the elevation of the turning point of vertical curves

It is not important for a vertical curve to be parabolic or circular in the determination of the turning point. Calculations are made using the general formulas obtained by the geometrical relations. If F is the elevation difference between the turning point and VPC, $x_{1,}$ and y_1 are the abscissa and ordinate of the turning point with respect to the curve start, $TP(DN)_{km}$ is the kilometer of the turning point, and $TP(DN)_{elv}$ is the elevation of the turning point,

$TP(DN)_{km} = T1_{km} + x_1$ $TP(DN)_{elv} = T1_{elv} + F$

can be written



$y=K.x^2$ general equation can be written in terms of the coordinates of the turning point (x_1, y_1) as, $y_1 = K.x_1^2$

If K=(G/2L) is written here, the equation becomes,

$$y_1 = \frac{G}{2L}.x_1^2$$

F, the difference between the turning point and PVC can be written as:

$$F = g_1.x_1 \quad y_1$$

If this equation is written in terms of x_1,

$$F = g_1.x_1 \quad \frac{G}{2L}.x_1^2$$

the equation is obtained. For F to be a maximum, its derivative has to be zero so,

$$\frac{dF}{dx_1} = g_1 \quad 2x_1.\frac{G}{2L} = 0$$

If x_1 is taken, $x_1 = L.\frac{g_1}{G}$

is obtained. If this term is written on the equation above, F becomes:

$$F = g_1.L.\frac{g_1}{G} \quad \frac{G}{2L}. \; L.\frac{g_1}{G}^{\;2}$$

by making necessary simplifications,

$$F = L.\frac{g_1^2}{G} \quad \frac{G}{2L}.\frac{L^2.g_1^2}{G^2} = L.\frac{g_1^2}{G} \quad \frac{1}{2}.L.\frac{g_1^2}{G} = L.\frac{g_1^2}{2.G}$$

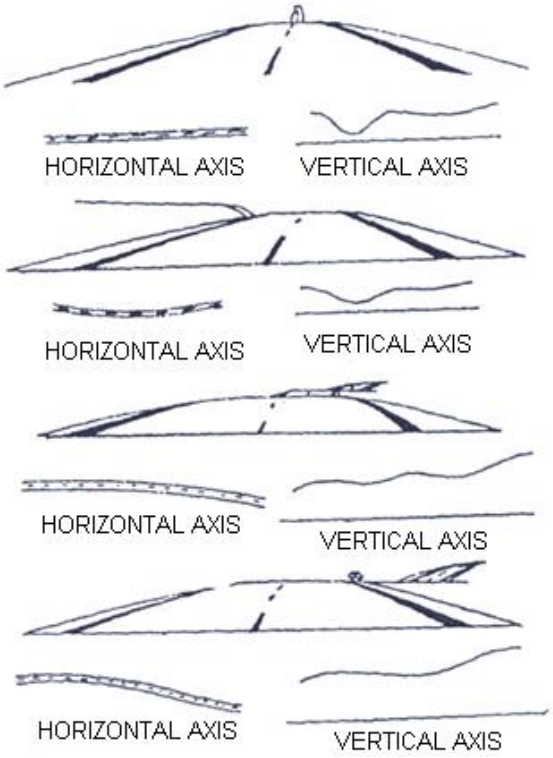
is obtained. Finally, The kilometer and the elevation of the turning point can be written as:

$$TP(DN)_{km} = T1_{km} + \frac{g_1 \quad L}{G} \qquad TP(DN)_{elv} = T1_{elv} + \frac{g_1^2 \quad L}{2G}$$

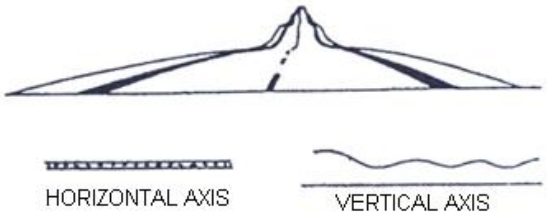
8.2.6 General rules for vertical axis design

On vertical axis design, some general criteria have to be taken into account in addition to ones already stated:

1. Instead of often changing the grades or planning short-distance grades, low grades compatible with the land topography should be selected for highway safety.
2. "Concealed-descending" type profiles have to be avoided. Usually, these types of profiles are seen where the vertical axis follows the undulated natural landline on long horizontal alignments. The profiles shown in the figure are undesirable situations aesthetically, and they cause driving difficulty. The driver willing to overtake a vehicle can be mistaken by thinking the opposite lane is empty after ascending. The driver is unaware of a vehicle approaching concealed behind a hill. These types of profiles can be corrected by horizontal curves or by using high cuts or fills and applying slowly changing grades.



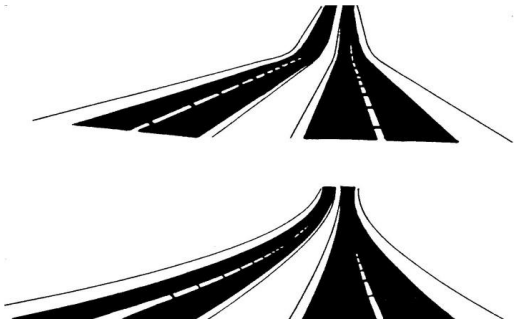
3. Persistently grade-changing (undulaced) profiles have to be avoided. The undulated profiles shown in the figure would cause heavy vehicles to move faster and thus may cause dangers.



4. As shown in the figure, a short alignment should be avoided between two same-sided vertical curves.



5. A short sag vertical curve should not be placed between two vertical alignments. As seen in the figure, the illumination length of the headlamp lights prevents the curve from being seen and endangers driving safety during the night drive. In this situation, the length of the sag vertical curve has to be increased to repair the optical brokenness and to increase highway safety. Determination of the curve lengths using the necessary K_{min} coefficient usually prevents these drawbacks.



6. Long descending profiles have to be avoided. If there isn't an ascending slope after a long decline, especially the overaccelerating heavy vehicles, it creates a danger to traffic.

7. On long ascending profiles, steeper grades have to be placed on lower elevations, and through the end of the climb, the grade has to be decreased, or instead of a constantly ascending grade, a broken vertical axis using short-leaning grades has to be designed.

8. On places where sufficient drainage cannot be provided at cut cross-sections, sag vertical curves have to be avoided.

9. Due to insufficient drainage, problems like hydroplane and friction resistance increase the accident risk, so the grade should not be lower than 0,5% on bouldered highways.

8.2.7 Compatibility of the horizontal and the vertical axis

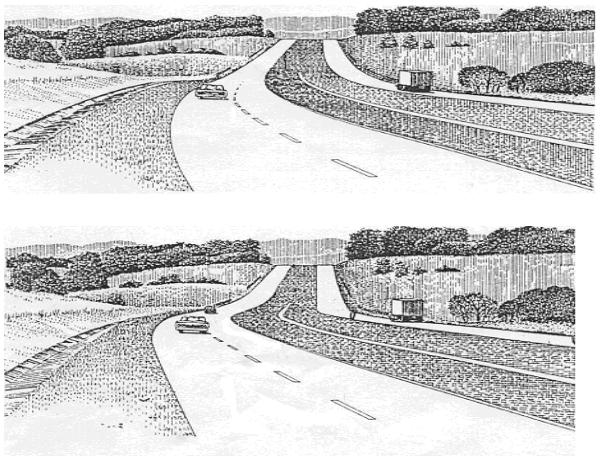
Horizontal and vertical axes are two of the main components of a highway, so they have to be designed carefully. Correcting the axis deformations after the construction phase is hard and expensive. Horizontal and vertical axis designs cannot be considered separately. The most economical way to provide aesthetics, in addition to the safe and steady traffic flow, is to determine whether these axes are compatible with each other in the beginning. The subjects that have to be taken into account on the compatibility of the horizontal and the vertical axis can be listed as:

1. Horizontal curve radii and longitudinal grades have to be compatible for stable and compatible horizontal and vertical geometry. Alignments or curves with high radii should not combine with steep and long grades; leaning grades should not combine with curves with low radii.

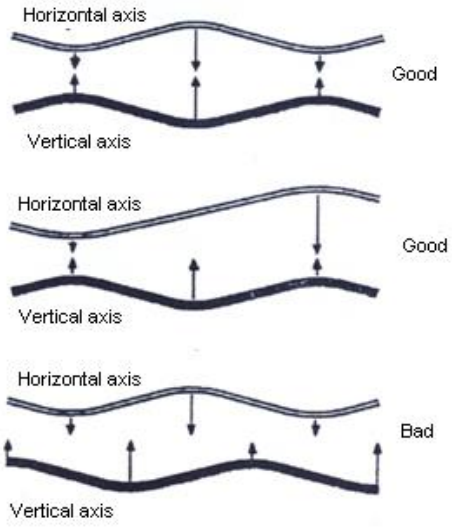
2. The effect of overlapping and/or coinciding horizontal and vertical curves on the traffic has to be studied. For overlapping and/or coinciding conditions, the following rules have to be taken into account for safety, comfort, and aesthetics:

    - At the beginning or at the top of a crest vertical curve, a sharp horizontal curve should not be used. Especially during the night drive, drivers may fail to recognize the horizontal change on the axis, which will increase the accident risk.
    - At the beginning or at the bottom of a sag vertical curve, a sharp horizontal curve should not be used. Otherwise, the driver may fail to see the section in front, increasing the accident risk.
    - A horizontal curve should never be placed after a steep vertical axis. In this case, the drivers fail to see forward during the night drive, and heavy vehicles may get out of the way due to their increasing velocity at the beginning of a sharp vertical curve.
    - If the horizontal and the vertical curves have to intersect, their length should be equal, and if possible, their start and end points should be the same; in other words, their intersection points should overlap.
    -

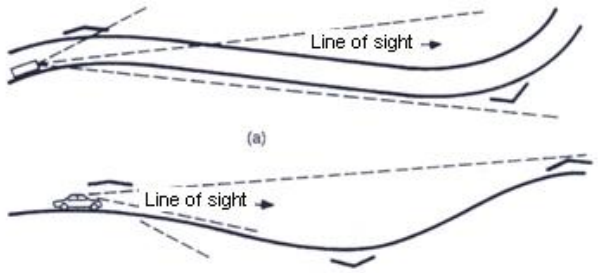
3. If a vertical curve has to be placed in a horizontal curve, provided that its effect on highway safety is analyzed, overlapping horizontal and vertical curves usually have a better sight. However, combining horizontal and vertical curves may result in undesirable situations. As seen in the figure, placing a short vertical curve at the end of a long horizontal curve gives a broken appearance to the highway. This undesirable situation can be resolved by increasing the vertical curve length.



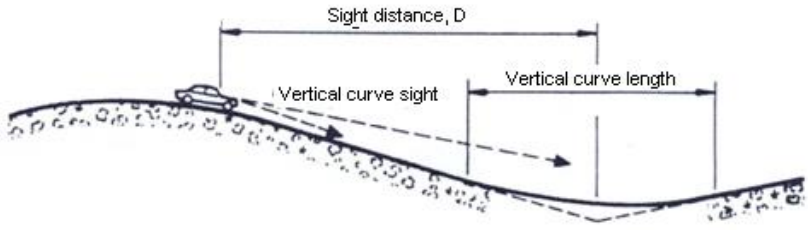
4. Desirable and undesirable situations of consequent horizontal and vertical curves are seen in the figure.



5. Horizontal and vertical axes should not be constituted of numerous same-direction curves. The visible distance of the highway should be designed by using 2 horizontal curves or 3 vertical curves at most, as seen in the figure. If there are more curves than the given value, the driver fails to recognize distance due to the lack of concentration. This condition is important in both day and night driving due to highway safety.

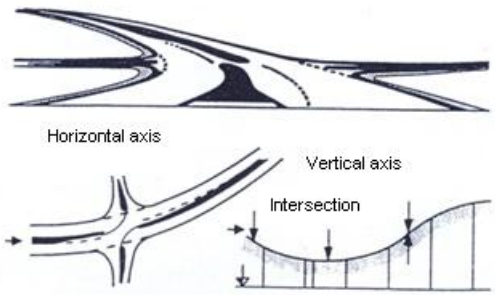


6. Especially during the night drive, the visibility of the horizontal and vertical curves has to be provided within the headlamp illumination distance. To provide the visibility of sag vertical curves, the vertical curve length has to be equal to the sight distance, or at least 0,6 times the sight distance. The horizontal curve length (in meters) has to be thrice the project velocity, preferably twice that amount. This statement shows some differences with the minimum curve lengths given in the previous lectures. However, the curve lengths given in the previous section show the minimum curve lengths with respect to performance and safety; the sight criterion may need some higher standards under some situations.



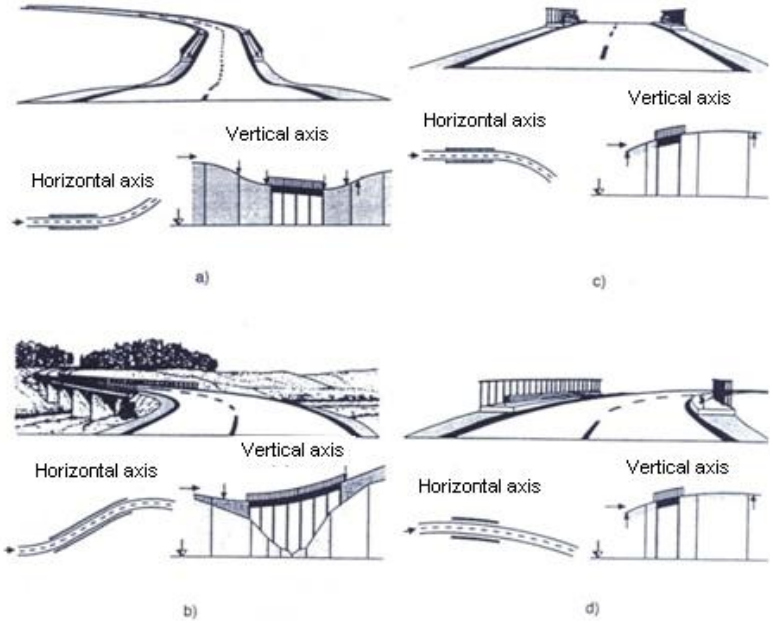
7. On intersection approaches, due to the deceleration and stopping needs of vehicles, horizontal and vertical curves have to be designed with high radii, and the necessary sight distance must be provided.

8. Intersections should be arranged on sag vertical curves, if possible, due to their visibility. The visibility of the intersection shown in the figure can be increased by proper measures like vegetation, illumination, and traffic signs. The necessary sight distance has to be provided at the intersections.



9. On bridge passes, compatibility of the horizontal and the vertical axes carries great importance, so the following matters have to be taken into account:

- The design of the engineering structures like bridges has to be compatible with the horizontal and the vertical axis, as seen in the figure b. Also, straight and flat locations like Figure A have to be avoided.
- For drivers to recognize the conditions like lateral wind effect and slippery pavement and adapt themselves to the changing conditions, bridges have to be visible.
- On bridges to be placed on the crest vertical curves, designs that prevent the visibility of the horizontal curve at the end of the vertical curve have to be avoided (Figure C). In this case, to provide the sight, the bridge should be placed in a curve.



PROBLEMS

1) A (2x1) lane stateway is to be designed for the project velocity of 90 km/h. On the first section of the profile, the red line grades are g_1 = 4.5% and g_2 = 2.5%, respectively. These lines intersect at a point with the kilometer (1+500) and the elevation of 145,0. Also the trucks can reach a velocity of 60 km/h at most, due to the grade on this section.

a) Calculate the length of the parabolic vertical curve to be placed on this highway section.

b) Taking L=350 m, find the kilometer and elevations of T_1, T_2, bisector and the turning point of the parabolic vertical curve.

c) If this highway have been designed as (2x2) lane, divided, would the length of the veritcal curve change ?

Given: $d_i = 0{,}2.\ V_i + 8$, $S > L \rightarrow L = 2.S-(4.4 / G)$, $S < L \rightarrow L = G.S^2/4.4$,

$f = 0{,}30$, t_r = 1 sn $TP(DN)_{km} = T1_{km} + (g_1.L)/G$, $TP(DN)_{elv} = T1_{elv} + (g_1^2.L)/(2.G)$

2) On a highway section having the red line grades g_1=-4%, g_2=2%, g_3=-2% respectively, $T1_{elv}$ =350 m, $T1_{km}$ =1+000, $T2_{elv}$ =365 m. A distance of 150 m. should be left between the T_2 of the first curve and T_2 of the second curve. Also there's a ratio of $t_1 = 1{,}5 * t_2$ between the tangents of the two vertical curves.

a) Calculate the lengths and the radii of the both vertical curves.

b) Find the kilometers of the bisector, T_2 and turning point of the first curve.

c) Find the red elevations of the T_1, bisector and turning point of the second curve.

3) Two red lines having grades of g_1 = 0,055 and g_2 = - 0,025 intersect at a PVI on a highway designed for a project velocity of 80 km/h. The distance of PVI to the start (2+400) m, and the elevation is 185 m. A circular vertical curve, having a radius of R =5000 m is to be placed between this two red lines.

a) Calculate the elevation and kilometers of T_1, P, bisector and turning point of the vertical curve.

b) Calculate the red elevations taken with 50 m intervals from the start of the vertical curve using the table given below.

Cross Section	**Km**	**x**	**$r = g_1 * x$**	$T1_{elv}$ **r**	**$y = K * x^2$**	$T1_{elv}$ **r y**

4) On a (2x2) lane divided stateway, designed for a project velocity of 80 km/h, a parabolic vertical curve is going to be placed between two red lines having the grade of g_1 = -3% and g_2 = 5% and intersecting at P with the red elevation of 139,5 m, kilometer of (3+200). Calculate length and elevation and kilometers of its tangets and bisector and complete the table below.

Given:

$d_i = 0{,}2 * V_i + 6$, $(V_1 - V_2) = 20$ km/h , t_r = 1 sec, f=0,40 ,

$S > L \rightarrow L = 2.S - (1{,}22 + 0{,}035.S)/G$,

$S < L \rightarrow L = (G.S^2)/(1{,}22 + 0{,}035.S)$

Cross-section	**Km**	**x**	**$r = g_1.x$**	$T1_{elv} \pm r$	**$y = K.x^2$**	$T1_{elv} \pm r \pm y$
T_1						
150	3+150					
151	3+175					
B	3+200					
T_2						